

2a

OSTEOARTHRITIS	RHEUMATOID ARTHRITIS
1.It is a degenerative disorder of the joint	It is a chronic inflammatory disease of the joint
2.Caused by a combination of many pathological both infective and degenerative e.g. ageing, obesity, trauma etc.	It is linked to autoimmune reaction
3.Systemic symptoms do not occur i.e. affects only the articular cartilage and the bone	Systemic symptoms may occur i.e. can affect other organs like heart, lung
4.Commonly affects the weight bearing joints e.g. hip, knee	Affects the small joints of the hands, feet, wrist, elbow etc.
5.May be hereditary	It is not hereditary
6.Deformity is not as pronounced as in rheumatoid	Rheumatoid factor in the blood
7.No rheumatoid factor in the blood	Deformity is more pronounced
8.No treatment halts the degenerative process	Treatment gives complete relief
9.Sinovectomy is not done	10.Sinovectomy is done
10.Absence of nodules	Presence of nodules

(1 mark each for any 4 pairs of points= 4 marks)

2b

PATHOPHYSIOLOGY OF ACUTE OSTEOARTHRITIS

- The articular cartilage of the bones becomes thin and worn and gradually break down leaving the bone exposed.
- There are outgrowths from the joint margin leading to thickening of the ends of the bones and the protruding ridges and spurs (osteophytes) .
- There is impairment of joint movement.
- The spurs may break off or become loose in the joint causing further impairment in the joint movement.
- This will lead to pain.

2c. NURSING CARE PLAN OF MADAM AGNES WITH OSTEOARTHRITIS

S/ No	Nursing Diagnoses	Objectives	Nursing Action	Scientific Principle/Rationale	Evaluation
1	Pain related to the disease process	Patient will verbalise less pain within 1-4 hours of treatment	1)Admit patient into an orthopaedic bed (with firm base or mattress) 2)Put patient on bed rest 3)Allow patient to assume any position of comfort	1) This maintains proper body alignment 2) Bed rest reduces activity level and unnecessary movement of the affected area thereby reduces	Patient verbalised less pain within 1-4 hours of treatment.



			4)Apply hot compress to the affected joints 5)Place patient on a graduated passive active exercise 6)Administrator prescribed analgesics and anti-inflammatory drugs e.g. A.S.A, Piroxican, Diclofenac, Tramadol, Ibuprofen	pain 3)This promotes patient's patient comfort and further reduces pain 4) Hot compress improves blood circulation to the affected areas and soothes the nerve ending. 5)This helps to maintain muscle tones and ensures gradual reform of normal movement of the part . 6)Analgesics block pain perception while the anti-inflammatory reduces the inflammatory process.	
2.	Impaired physical inability related to the disease process	Patient will move the joint with relative ease within 12-24 hours of treatment	1)Continue with the application of hot and cold compress 2)Encourage passive/active exercise 3)Health educate the patient on the need for exercise 4)Place patient on nourishing diet 5)Advise patient on need to reduce weight 6)Continue with the administration of prescribed analgesic and anti-inflammatory drugs	1) This further soothes and promote healing of the affected part. 2) Exercises maintains and strengthens the muscle tone and the power of the limb. 3) Education promotes confidence in the acceptance of care and ensure compliance 4) Nourishing diet provides energy for the exercises and help fight infection 5) This helps reduce patient weight and stress on the affected joint and promote	Patient was able to move the joints with relative ease within 12-24 hours of intervention



				healing 6)This promotes healing and comfort	
3	Anxiety related to diseases process	Patient will be less anxious 2-6 hours of nursing care	1)Assess the level of anxiety 2)Establish a good rapport with the patient 3)Allow patient to verbalise her concerns and ask questions 4)Reassure the patient and family based on her questions and concerns 5)Administer prescribed medications e.g. sedative, anxiolytic like diazepam.	1)Assessment gives baseline data for necessary action 2)This enhances good interpersonal relationship between nurse and the patient 3)This allows for identification and correction of areas of misconception 4)Reassurance allays anxiety 5)This calm patients, promotes rest and improves the general well being of the patient.	Patient was less anxious within 3 hours of nursing actions as evidenced by her participation in her care.

(3 marks each for any correct plan)

OTHERS

- Body image disturbance
- Activity intolerance
- Ineffective individual coping
- Self care deficit
- Sleep pattern disturbance

½ mark – diagnosis, ¼ mark –

objectives, 1 mark –Action(4points)

(½ mark should be awarded to any student that developed other nursing diagnoses listed below)

2d. COMPLICATIONS OF OSTEOARTHRITIS

1. Flexion Contractures
2. Deformity
3. Ankylosis
4. Bony Cysts
5. Gross bony overgrowths
6. Nerve root compression



7. Cauda equine syndrome
8. Subluxation (Sprain dislocation)
9. Central Cord Syndrome (with cervical spine osteoarthritis)

3a. Disaster can be defined as a sudden, unexpected event or misfortune that is natural or manmade and which causes great destruction of lives and properties and affects normal activities of the community.

OR

Disaster can be defined as an accident that causes great damage to lives and properties and sufficient enough to require extra-ordinary mobilization of resources beyond those readily available within the community.

(¼ mark for any 4 underlined words= 1mark)

3b

TYPES OF DISASTER

- i) Natural Disaster: This results from violence of nature's element related to weather or other geographical conditions.

Examples : Hurricanes, Earthquakes, Flood/Erosion, Drought, Thunder blast, Landslides, Extremes of temperature, Volcano Eruptions, Snow storms, Typhoons.

- ii) Man made Disaster: This is caused by man as a result of dubious/carelessness/negligence in his bid to carry out revenge or protection or in search of knowledge or research.

Examples : Automobile accident, Fire outbreak, Bomb blast, Terrorist action, Pipeline explosion, Civil disorder, Wars, Outbreak of infectious disease, Emission of toxic wastes, Industrial accidents, Poison e.g. drugs, food, chemicals, water etc.

(Identification of type = 1 mark each= 2marks)

(¼ mark for any 4 correct points in each type=1mark)

(Total = 4marks)

3c **MANAGEMENT OF DISASTER VICTIMS**

- I Rescue the victim
- ii Quick assessment using Triage
- iii Resuscitation of victims using ABC of trauma, disability and evaluation.
- Iv Ensure patient airway by correct positioning and suctioning.



- v Remove all tight clothing around neck, chest and waist.
- vi Do mouth to mouth resuscitation and external cardiac massage
- vii Control bleeding by applying pressure to the wound areas using emergency tourniquet such as scarf, handkerchief or hand towel.
- Viii Cover wound with a clean material as much as possible.
- ix Place unconscious patients in prone position or lying on their side.
- X Do not give oral fluid to unconscious or semiconscious victims
- Xi Do not try to arouse unconscious persons by slapping or shaking
- Xii Perform brief physical examination and assessment
- Xiii Give psychotherapy to victims and family members
- Xiiii Determine the areas of primary providers and refer.
- Xv Initiate and provide documentation of time, condition and initial treatment
- Xvi Initiate therapeutic or diagnostic interventions e.g. lab test,
- Xvii Call for assistance from appropriate agencies such as NEMA, Red Cross, Police etc.
- Xviii Prevention of disability through proper handling of victim e.g. Immobilization of fractures, Proper positioning
- Xix Rehabilitation of victims e.g. provision of alternative accommodation, feeding and clothing.
- Xx Handling of effects of disasters e.g. clearing of the corpses off the street to avoid further danger.

(½ marks for any correct point =10 marks)

3d

ROLE OF GOVERNMENT IN DISASTER PREPAREDNESS

- Government should provide motorable roads
- Provide equipment required for handling disasters e.g. vehicles.
- Setting up and proper funding of bodies for management of disasters including manpower training.
- Monitoring of geographical events, elections and other human activities that can result in disaster



- Issuing warning on impending disasters such as drought, hurricanes.
- Enacting laws on prevention of disasters e.g. use of seatbelts, drinking and driving and other dangerous practices
- Health education on causes and the evil effects of disaster
- Employ well trained man power

ROLE OF COMMUNITY IN DISASTER

- Maintenance of provided equipment
- Alertness and surveillance
- Reporting incidences promptly and unusual experiences
- Formation of disaster intervention team
- Providing succour to victims

Adherence to safety regulations

(½ mark for any 3 correct points in each role = 3marks)

4a

i. NATURAL RESERVOIR OF THE LASSA VIRUS

The natural reservoir of the Lassa virus is the rodent of the genus *Mastomys* commonly known as the “multimammate rat”. Even though the exact species that causes the disease is not exactly known, it is presumed to be caused by the species *natalensis*. (1 mark)

ii. CLINICAL MANIFESTATIONS OF LASSA FEVER

Signs and symptoms of Lassa fever typically occur 1-3 weeks after the patient comes into contact with the virus. These include:

- Fever
- Abdominal pain
- Retrosternal pain
- Vomiting
- Pharyngitis
- Cough
- Back pain
- Facial swelling
- Diarrhea
- Proteinuria
- Conjunctivitis
- General malaise



- Mucosal bleeding - Headache
- Diarrhea or constipation - Shock
- Respiratory distress,
- Neurological problems like hearing loss, tremors, and encephalitis.

(Any 6 correct points = 3 marks)

iii. MODE OF TRANSMISSION

The virus can be transmitted to humans in many different ways. *M. natalensis* live in and around houses, where they expel the virus in urine and feces. Consequently, the virus can be transmitted **directly, indirectly, or airborne**.

- Lassa virus can be transmitted directly if a person is exposed to the blood, urine, feces, or other body fluids from an infected person. It can also be transmitted by sexual contact. Transmission through breast milk has also been observed.
- Indirect transmission by means of a fomite can also occur. If people come across a contaminated object, they can become infected as well. In addition, people can contract the disease by contaminated food. For instance, in some parts of Africa, *M. natalensis* is used as a food source and so people can contract the disease by eating contaminated meat.
- Although airborne transmission is not a characteristic of arenaviruses, the Lassa virus can be transmitted by air particles contaminated with rodent excretions

(Any 3 correct points discussed = 3 marks)

iv. PREVENTION OF SPREAD OF LASSA VIRUS

- Primary transmission of the Lassa virus from its host to humans can be prevented by avoiding contact with *Mastomys* rodents, especially in the geographic regions where outbreaks occur. Trapping in and around homes can help reduce rodent populations.
- Putting food away in rodent-proof containers and keeping the home clean help to discourage rodents from entering homes.
- Using these rodents as a food source is not recommended.
- When caring for patients with Lassa fever, further transmission of the disease through person-to-person contact or nosocomial routes can be



avoided by taking preventive precautions against contact with patient secretions (barrier nursing methods). Such precautions include wearing protective clothing, such as masks, gloves, gowns, and goggles; using infection control measures, such as complete equipment sterilization; and isolating infected patients from contact with unprotected persons until the disease has run its course.

- Further educating people in high-risk areas about ways to decrease rodent populations in their homes will aid in the control and prevention of Lassa fever.

(Any correct 4 points discussed= 4 marks)

iv. COMPLICATIONS OF LASSA FEVER

- Deafness
- Spontaneous abortion (Each correct point = $\frac{1}{2}$ mark)

4b METHODS OF WASTE DISPOSAL

Landfill

The Landfill is the most popularly used method of waste disposal used today. This process of waste disposal focuses attention on burying the waste in the land. Landfills are found in all areas. It is a process used to eliminate the odors and dangers of waste before it is placed into the ground. It is

Incineration/Combustion

Incineration or combustion is a type of disposal method in which municipal solid wastes are burned at high temperatures so as to convert them into residue and gaseous products. The biggest advantage of this method is that it can reduce the volume of solid waste to 20 to 30 percent of the original volume, decreases the space they take up and reduce the stress on landfills. This process is also known as **thermal treatment** where solid waste materials are converted by Incinerators into heat, gas, steam and ash. Incineration is something that is very common in countries where landfill space is no longer available.

Recovery and Recycling

Resource recovery is the process of taking useful discarded items for a specific next use. These discarded items are then processed to extract or recover materials and resources or convert them to energy in the form of useable heat, electricity or fuel.



Recycling of waste material means taking the materials and transforming them into new products. This is a key concept in the modern waste minimization philosophy. It is about lessening the strain on the environment through minimizing the need to fully dispose (e.g. by incineration and causing air pollution) of the waste generated and reducing the need to introduce new raw materials into the environment and then having to dispose of them later.

Recycling is the third component of Reduce, Reuse and Recycle waste hierarchy. The idea behind recycling is to reduce energy usage, reduce volume of landfills, reduce air and water pollution, reduce greenhouse gas emissions and preserve natural resources for future use.

Composting

Composting is an easy and natural bio-degradation process that takes organic wastes i.e. remains of plants and garden and kitchen waste and turns into nutrient-rich food for plants. Composting, normally used for organic farming, occurs by allowing organic materials to sit in one place for months until microbes decompose it. Composting is one of the best methods of waste disposal as it can turn unsafe organic products into safe compost. On the other side, it is slow process and takes lot of space.

Plasma gasification

Plasma gasification is another form of waste management. Plasma is a primarily an electrically charged or a highly ionized gas. Lighting is one type of plasma which produces temperatures that exceed 12,600 °F. With this method of waste disposal, a vessel uses characteristic plasma torches operating at +10,000 °F which is creating a gasification zone till 3,000 °F for the conversion of solid or liquid wastes into a syngas.

During the treatment of solid waste by plasma gasification, the waste's molecular bonds are broken down as result of the intense heat in the vessels and the elemental components. Destruction of waste and dangerous materials is found. This form of waste disposal provides renewable energy and an assortment of other fantastic benefits.

Waste to Energy (Recover Energy)

Waste to Energy (WtE) process involves converting of non-recyclable waste items into useable heat, electricity, or fuel through a variety of processes. This type of source of energy is a renewable energy source as non-recyclable waste can be used over and over again to create energy. It can also help to reduce carbon emissions by offsetting the need for energy from fossil sources. It is the generation of energy in the form of heat or electricity from waste.

(Any 4 correct points discussed = 8 marks)



5a CHARACTERISTICS OF A PROFESSION AND THEIR APPLICATION TO NURSING

The following are the common characteristics of a profession:

- A unique, definite and essential service. It renders an essential social service.

Nursing is an essential service to individual and society especially in the health sector.

- An emphasis upon intellectual techniques in performing its services i.e. requires intellectual techniques to define problems, search for relevant data and formulate possible solutions and not mainly on physical techniques.

Nurses acquire intellectual skills during training and apply them in the care of the clients.

- A long period of specialized training i.e. requires a long period of preparation.

Nursing students attend schools of nursing for three years or universities for five years to acquire specialized knowledge and practical training.

- A broad range of autonomy for both the individual practitioners and the occupational group. They are free to exercise their own best judgment and to exercise independent skill individually. Also, the practitioners of a profession enjoy a broad measure of autonomy as a group.

Nurses have power to decide the qualifications for admission into nursing school, the ground for suspension or exclusion for practice. Also, nurses have the right to decide on the nursing care to be given to their clients.

- It demands continuous in service training of its members.

Nurses (especially Nursing and Midwifery Council of Nigeria) organize in service training for its members like seminars and workshops (e.g.

